Chapter 18, Problem 28P

Problem

Find the inverse Fourier transforms of:

(a) $\dfrac{\pi\delta(\omega)}{(5 + j\omega)(2 + j\omega)}$

(b) $\dfrac{10\delta(\omega + 2)}{j\omega(j\omega + 1)}$

(c) $\dfrac{20\delta(\omega - 1)}{(2 + j\omega)(3 + j\omega)}$

(d) $\dfrac{5\pi\delta(\omega)}{5 + j\omega} + \dfrac{5}{j\omega(5 + j\omega)}$

Step-by-step solution

Step 1 of 4

(a)

Consider the following Fourier transform:

$$F(\omega) = \frac{\pi\delta(\omega)}{(5 + j\omega)(2 + j\omega)}$$

Apply the inverse Laplace transform.

$$\begin{aligned} f(t) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega) e^{j\omega t}\, dt \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{\pi\delta(\omega) e^{j\omega t}}{(5 + j\omega)(2 + j\omega)}\, d\omega \\ &= \frac{1}{2\pi}\frac{\pi\delta(0) e^{j0t}}{(5 + j0)(2 + j0)} \\ &= \frac{1}{20} \end{aligned}$$

Therefore, the inverse Fourier transform of $\dfrac{\pi\delta(\omega)}{(5 + j\omega)(2 + j\omega)}$ is $\boxed{\dfrac{1}{20}}$.

Step 2 of 4

(b)

Consider the following Fourier transform:

$$F(\omega)=\frac{10\delta(\omega+2)}{(j\omega)(j\omega+1)}$$

Apply the inverse Laplace transform.

$$\begin{aligned}f(t)&=\frac{1}{2\pi}\int_{-\infty}^{\infty}F(\omega)e^{j\omega t}dt\\&=\frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{10\delta(\omega+2)}{(j\omega)(j\omega+1)}e^{j\omega t}d\omega\\&=\frac{1}{2\pi}\frac{10\delta(0)e^{-j2t}}{(-j2)(-j2+1)}\\&=\frac{j5e^{-j2t}}{2\pi(1-j2)}\end{aligned}$$

$$=\frac{(-2+j)e^{-j2t}}{2\pi}$$

Therefore, the inverse Fourier transform of $\frac{10\delta(\omega+2)}{(j\omega)(j\omega+1)}$ is $\boxed{\frac{(-2+j)e^{-j2t}}{2\pi}}$.

Step 3 of 4

(c)

Consider the following Fourier transform:

$$F(\omega)=\frac{20\delta(\omega-1)}{(2+j\omega)(3+j\omega)}$$

Apply the inverse Laplace transform.

$$\begin{aligned}f(t)&=\frac{1}{2\pi}\int_{-\infty}^{\infty}F(\omega)e^{j\omega t}dt\\&=\frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{20\delta(\omega-1)}{(2+j\omega)(3+j\omega)}e^{j\omega t}d\omega\\&=\frac{1}{2\pi}\frac{20\delta(0)e^{jt}}{(2+j)(3+j)}\\&=\frac{20e^{jt}}{2\pi(5+5j)}\end{aligned}$$

$$=\frac{(1-j)e^{jt}}{\pi}$$

Therefore, the inverse Fourier transform of $\frac{20\delta(\omega-1)}{(2+j\omega)(3+j\omega)}$ is $\boxed{=\frac{(1-j)e^{jt}}{\pi}}$.

Step 4 of 4

(d)

Consider the following Fourier transform:

$$F(\omega)=\frac{5\pi\delta(\omega)}{(5+j\omega)}+\frac{5}{j\omega(5+j\omega)}$$

Apply the inverse Laplace transform.

$$\begin{aligned}
f(t)&=\frac{1}{2\pi}\int_{-\infty}^{\infty}F(\omega)e^{j\omega t}dt\\
&=\frac{1}{2\pi}\int_{-\infty}^{\infty}\left[\frac{5\pi\delta(\omega)}{(5+j\omega)}+\frac{5}{j\omega(5+j\omega)}\right]e^{j\omega t}d\omega\\
&=\frac{1}{2\pi}\left[\frac{5\pi\delta(0)}{(5+j0)}\right]e^{j0t}+\int_{-\infty}^{\infty}\frac{1}{2\pi}\left[\frac{(5+j\omega)-j\omega}{j\omega(5+j\omega)}\right]e^{j\omega t}d\omega\\
&=\frac{1}{2}+\int_{-\infty}^{\infty}\frac{1}{2\pi}\left[\frac{1}{j\omega}\right]e^{j\omega t}d\omega-\int_{-\infty}^{\infty}\frac{1}{2\pi}\left[\frac{1}{(5+j\omega)}\right]e^{j\omega t}d\omega
\end{aligned}$$

$$=\frac{1}{2}+\frac{1}{2}\operatorname{sgn}(t)-e^{-5t}$$

Therefore, the inverse Fourier transform of $\frac{5\pi\delta(\omega)}{(5+j\omega)}+\frac{5}{j\omega(5+j\omega)}$ is,

$$\boxed{=\frac{1}{2}+\frac{1}{2}\operatorname{sgn}(t)-e^{-5t}}.$$